

HADOOP commands

Step 1: Create a Sample File

Open the terminal (localhost:4200) and create a file:

```
vi data.txt
```

Add some lines:

```
Big data technologies are important  
Hadoop stores large datasets  
MapReduce processes distributed data
```

Save using:

```
Esc → :wq
```

Step 2: Create Directory in HDFS

```
hdfs dfs -mkdir /blockdemo
```

Step 3: Upload File to HDFS

```
hdfs dfs -put data.txt /blockdemo
```

Step 4: Verify File in HDFS

```
hdfs dfs -ls /blockdemo
```

Example output:

```
/blockdemo/data.txt
```

Step 5: Check File Block Information

Run:

```
hdfs fsck /blockdemo/data.txt -files -blocks -locations
```

You will see output like:

```
Block replica on datanode  
Block ID: blk_1073741825
```

Namenode :

With respect to namenode

- `hdfs dfsadmin -report`
 - NameNode status (live)
 - Configured capacity, used space
 - Live DataNodes / Dead DataNodes
- 👉 This is the most commonly used verification command.
-

2. Check if NameNode is running (process level)

`jps`

Output should include:

NameNode

DataNode

SecondaryNameNode

3. Get detailed NameNode configuration

`hdfs getconf -confKey fs.defaultFS`

Example output:

`hdfs://sandbox-hdp.hortonworks.com:8020`

This shows the **NameNode URI (host + port)**